

CONSTRUCTION AND ANALYSIS OF THE MODULAR CURVE $X_0(2)$

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ABSTRACT. We construct the modular curve $X_0(2)$ as a compact Riemann surface. We find a fundamental domain for $\Gamma_0(2)$, show that $X_0(p)$ has exactly two cusps for every prime p , and prove that $X_0(2)$ is compact of genus 0. We then write down an explicit model of $X_0(2)$ over $\mathbb{Q}$. It describes the j -invariants of elliptic curves with a rational 2-isogeny, and we illustrate it with an explicit example.

1. CONSTRUCTION OF $Y_0(2)$

In this section we construct the modular curve $Y_0(2)$ as a Riemann surface over $\mathbb{C}$. The group $\mathrm{SL}_2(\mathbb{Z})$ acts on the upper half plane $\mathbb{H} = \{\tau \in \mathbb{C} : \mathrm{Im}(\tau) > 0\}$ by fractional linear transformations: for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ and $\tau \in \mathbb{H}$,

$$\gamma(\tau) = \frac{a\tau + b}{c\tau + d}.$$

We define $Y(1)$ to be the quotient of $\mathbb{H}$ by this action,

$$Y(1) = \mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H},$$

which is a Riemann surface but is not compact. Now consider the congruence subgroup

$$\Gamma_0(2) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : c \equiv 0 \pmod{2} \right\}.$$

We define $Y_0(2)$ as the further quotient

$$Y_0(2) = \Gamma_0(2) \backslash \mathbb{H},$$

which, once again, is a Riemann surface but is not compact.

2. A FUNDAMENTAL DOMAIN FOR $\Gamma_0(2)$

We can find a fundamental domain for $\Gamma_0(2)$ using the standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$,

$$\mathcal{F} = \left\{ \tau \in \mathbb{H} : |\mathrm{Re}(\tau)| \leq \frac{1}{2}, |\tau| \geq 1 \right\}.$$

Since $\Gamma_0(2)$ is a proper subgroup of $\mathrm{SL}_2(\mathbb{Z})$, fewer points of $\mathbb{H}$ are identified, and the fundamental domain grows. The extra pieces are determined by coset representatives for $\Gamma_0(2) \backslash \mathrm{SL}_2(\mathbb{Z})$. Let

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

which generate $\mathrm{SL}_2(\mathbb{Z})$.

Lemma 2.1. *The index of $\Gamma_0(2)$ in $\mathrm{SL}_2(\mathbb{Z})$ is 3, and a set of right coset representatives for $\Gamma_0(2)$ in $\mathrm{SL}_2(\mathbb{Z})$ is*

$$\gamma_1 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \gamma_2 = S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \gamma_3 = ST = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}.$$

Proof. Send $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ to the point $(c : d)$ of the projective line $\mathbb{P}^1(\mathbb{F}_2)$, reducing modulo 2. This is well defined because $\gcd(c, d) = 1$. Two matrices γ, γ' lie in the same right coset $\Gamma_0(2)\gamma$ exactly when $\gamma'\gamma^{-1}$ has even lower-left entry, which happens exactly when their bottom rows agree in $\mathbb{P}^1(\mathbb{F}_2)$. Since every point of $\mathbb{P}^1(\mathbb{F}_2)$ is hit, this gives a bijection $\Gamma_0(2)\backslash\mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathbb{P}^1(\mathbb{F}_2)$, so the index is $|\mathbb{P}^1(\mathbb{F}_2)| = 3$. The bottom rows of I , S and ST are $(0 : 1)$, $(1 : 0)$ and $(1 : 1)$, which are the three distinct points of $\mathbb{P}^1(\mathbb{F}_2)$. $\square$

Proposition 2.2. *The set*

$$D = \mathcal{F} \cup S(\mathcal{F}) \cup (ST)(\mathcal{F})$$

is a fundamental domain for $\Gamma_0(2)$.

Proof. Let $\tau \in \mathbb{H}$. Since $\mathcal{F}$ is a fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$, there is a $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ with $\tau' = \gamma(\tau) \in \mathcal{F}$. By lemma 2.1 we can write $\gamma^{-1} = g\gamma_i$ with $g \in \Gamma_0(2)$, and then $g^{-1}(\tau) = \gamma_i(\tau') \in \gamma_i(\mathcal{F})$. So every point of $\mathbb{H}$ is $\Gamma_0(2)$ -equivalent to a point of D . Conversely, suppose two points of D in the interiors of $\gamma_i(\mathcal{F})$ and $\gamma_k(\mathcal{F})$ are $\Gamma_0(2)$ -equivalent via some g . Pulling both back to $\mathcal{F}$ gives two interior points of $\mathcal{F}$ related by $\gamma_k^{-1}g\gamma_i$. Since interior points of $\mathcal{F}$ are not equivalent under $\mathrm{SL}_2(\mathbb{Z})$ unless they are equal (and are fixed only by $\pm I$), we get $\gamma_k^{-1}g\gamma_i = \pm I$. Hence $\Gamma_0(2)\gamma_i = \Gamma_0(2)\gamma_k$, so $i = k$ and the two points coincide. $\square$

Concretely, $S(\tau) = -1/\tau$ and $(ST)(\tau) = -1/(\tau + 1)$. Both pieces $S(\mathcal{F})$ and $(ST)(\mathcal{F})$ have a cusp at 0, and together they form a region of width 2 there. Figure 1 shows the picture. Note that $T \in \Gamma_0(2)$, so every point of $\mathbb{H}$ can be translated into the strip $|\mathrm{Re}(\tau)| \leq \frac{1}{2}$. The map T also identifies the upper portions of the two vertical sides of D .

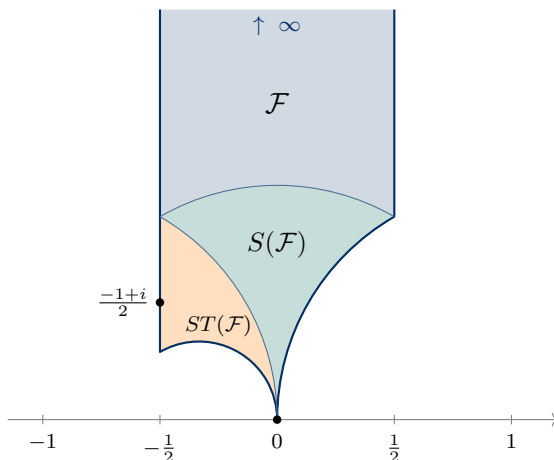


FIGURE 1. The fundamental domain $D = \mathcal{F} \cup S(\mathcal{F}) \cup ST(\mathcal{F})$ for $\Gamma_0(2)$, with its two cusps ∞ and 0 and the elliptic point $\frac{-1+i}{2}$ of period 2.

3. CONSTRUCTION OF $X_0(2)$

To compactify $Y_0(2)$, let $\mathbb{H}^* = \mathbb{H} \cup \mathbb{Q} \cup \{\infty\}$ and define the extended quotient

$$X_0(2) = \Gamma_0(2)\backslash\mathbb{H}^*.$$

More generally, for a positive integer N we write $X_0(N) = \Gamma_0(N)\backslash\mathbb{H}^*$. We want to show that $X_0(2)$ is a compact Riemann surface.

3.1. Cusps. The points added in passing from $Y_0(N)$ to $X_0(N)$ are the *cusps*. They correspond to the $\Gamma_0(N)$ -equivalence classes of $\mathbb{Q} \cup \{\infty\}$.

Proposition 3.1. *Let p be a prime. The modular curve $X_0(p)$ has exactly two cusps, represented by ∞ and 0 . More precisely, a rational number $\frac{m}{n}$ in lowest terms is $\Gamma_0(p)$ -equivalent to ∞ if $p \nmid n$, and to 0 if $p \nmid n$.*

Proof. The cusps 0 and ∞ are inequivalent. A matrix $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ acts on ∞ by $\gamma(\infty) = a/c$. If $\gamma(\infty) = 0$, then $a = 0$, so $\det \gamma = -bc = 1$ forces $c = \pm 1$. Then $p \nmid c$, so $\gamma \notin \Gamma_0(p)$. Hence 0 and ∞ represent distinct cusps.

Every rational number is equivalent to 0 or ∞ . Let $\frac{m}{n} \in \mathbb{Q}$ with $\gcd(m, n) = 1$.

Suppose first that $p \nmid n$. Then $\gcd(pm, n) = 1$, so there are $x, y \in \mathbb{Z}$ with $pmx + ny = 1$. The matrix

$$\gamma = \begin{pmatrix} n & -m \\ px & y \end{pmatrix}$$

has determinant $ny + pmx = 1$ and lower-left entry divisible by p , so $\gamma \in \Gamma_0(p)$, and

$$\gamma\left(\frac{m}{n}\right) = \frac{n \cdot \frac{m}{n} - m}{px \cdot \frac{m}{n} + y} = \frac{mn - mn}{pmx + ny} = \frac{0}{1} = 0.$$

Suppose instead that $p \mid n$. Then $p \nmid m$, and there are $x, y \in \mathbb{Z}$ with $-mx - ny = 1$. The matrix

$$\gamma = \begin{pmatrix} x & y \\ n & -m \end{pmatrix}$$

has determinant $-mx - ny = 1$ and lower-left entry $n \equiv 0 \pmod{p}$, so $\gamma \in \Gamma_0(p)$. Its denominator vanishes at $\frac{m}{n}$ while its numerator does not. Thus every element of $\mathbb{Q} \cup \{\infty\}$ is $\Gamma_0(p)$ -equivalent to 0 or ∞ , and $X_0(p)$ has exactly two cusps. $\square$

3.2. Topology and compactness. We now put a topology on $\mathbb{H}^*$ that extends the usual topology on $\mathbb{H}$. For $M > 0$, let

$$\mathcal{N}_M = \{\tau \in \mathbb{H} : \mathrm{Im}(\tau) > M\}.$$

We take as a base for the topology on $\mathbb{H}^*$ the usual open subsets of $\mathbb{H}$ together with the sets

$$\alpha(\mathcal{N}_M \cup \{\infty\}), \quad M > 0, \alpha \in \mathrm{SL}_2(\mathbb{Z}),$$

which serve as the neighborhoods of the cusps. For $\alpha(\infty) = s \in \mathbb{Q}$, such a set is an open disc in $\mathbb{H}$ tangent to $\mathbb{R}$ at s , together with the point s . With this topology every element of $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathbb{H}^*$ as a homeomorphism, and we give $X_0(2)$ the quotient topology.

Lemma 3.2. *The set $\mathcal{F}^* = \mathcal{F} \cup \{\infty\}$ is compact in $\mathbb{H}^*$.*

Proof. Let $\{U_\alpha\}$ be an open cover of $\mathcal{F}^*$. Some member U_{α_∞} contains ∞ , and therefore contains a basic neighborhood $\mathcal{N}_M \cup \{\infty\}$ for some $M > 0$. What remains to be covered is

$$K_M = \{\tau \in \mathcal{F} : \mathrm{Im}(\tau) \leq M\}.$$

This set is closed and bounded in $\mathbb{C}$, since $|\mathrm{Re} \tau| \leq \frac{1}{2}$ and $\frac{\sqrt{3}}{2} \leq \mathrm{Im} \tau \leq M$, and it lies inside $\mathbb{H}$. So K_M is compact, and finitely many sets $V_1, \dots, V_N$ from the cover suffice to cover it. Together with $V_0 = U_{\alpha_\infty}$, the sets $V_0, V_1, \dots, V_N$ form a finite subcover of $\mathcal{F}^*$. $\square$

Proposition 3.3. *The modular curve $X_0(2)$ is compact.*

Proof. Every point of $\mathbb{H}$ is $\mathrm{SL}_2(\mathbb{Z})$ -equivalent to a point of $\mathcal{F}$, and every cusp is of the form $\alpha(\infty)$ with $\alpha \in \mathrm{SL}_2(\mathbb{Z})$. Hence $\mathbb{H}^* = \mathrm{SL}_2(\mathbb{Z})(\mathcal{F}^*)$. By lemma 2.1, $\mathrm{SL}_2(\mathbb{Z}) = \bigcup_{i=1}^3 \Gamma_0(2)\gamma_i$, so

$$\mathbb{H}^* = \bigcup_{i=1}^3 \Gamma_0(2)\gamma_i(\mathcal{F}^*).$$

Let $\pi : \mathbb{H}^* \rightarrow X_0(2)$ be the quotient map. Since π identifies $\Gamma_0(2)$ -orbits,

$$X_0(2) = \pi(\mathbb{H}^*) = \bigcup_{i=1}^3 \pi(\gamma_i(\mathcal{F}^*)).$$

Each γ_i is a homeomorphism and π is continuous, so each $\pi(\gamma_i(\mathcal{F}^*))$ is compact by lemma 3.2. A finite union of compact sets is compact. $\square$

Remark 3.4. More generally, $X_0(N)$ is a connected, compact, Hausdorff Riemann surface for every positive integer N , and the same holds for $X(\Gamma)$ for any congruence subgroup Γ ; see [1, Chapter 2]. The noncuspidal points of $X_0(N)$ parametrize isomorphism classes of pairs (E, C) , where E is an elliptic curve over $\mathbb{C}$ and $C \subset E$ is a cyclic subgroup of order N . Equivalently, they parametrize cyclic N -isogenies $E \rightarrow E/C$.

3.3. The genus of $X_0(2)$. To compute the genus of $X_0(2)$ we use the following theorem.

Theorem 3.5 ([1, Theorem 3.1.1]). *Let Γ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Let $f : X(\Gamma) \rightarrow X(1)$ be the natural projection and let d denote its degree. Let ε_2 and ε_3 denote the numbers of elliptic points of period 2 and 3 in $X(\Gamma)$, and let ε_∞ denote the number of cusps of $X(\Gamma)$. Then the genus of $X(\Gamma)$ is*

$$g = 1 + \frac{d}{12} - \frac{\varepsilon_2}{4} - \frac{\varepsilon_3}{3} - \frac{\varepsilon_\infty}{2}.$$

For $\Gamma_0(2)$ the data are as follows.

- $d = 3$. Since $-I \in \Gamma_0(2)$, the degree equals the index from lemma 2.1.
- $\varepsilon_\infty = 2$, by proposition 3.1.
- $\varepsilon_2 = 1$. The point $\frac{-1+i}{2} = (ST)(i)$ is fixed by $\begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix} \in \Gamma_0(2)$. In general, ε_2 counts the solutions of $x^2 + 1 \equiv 0 \pmod{N}$, and there is exactly one for $N = 2$.
- $\varepsilon_3 = 0$. Here ε_3 counts the solutions of $x^2 + x + 1 \equiv 0 \pmod{N}$, and there are none modulo 2.

d	ε_2	ε_3	ε_∞	g
3	1	0	2	0

TABLE 1. Ramification data for $X_0(2) \rightarrow X(1)$.

Substituting these values gives

$$g = 1 + \frac{3}{12} - \frac{1}{4} - 0 - \frac{2}{2} = 0.$$

Corollary 3.6. *$X_0(2)$ has genus 0, and so $X_0(2) \cong \mathbb{P}^1(\mathbb{C})$ as Riemann surfaces.*

Proof. Pick a point $x \in X_0(2)$. By Riemann–Roch, a compact Riemann surface of genus 0 has $\ell(x) = \deg(x) - g + 1 = 2$. So there is a nonconstant meromorphic function t with a single simple pole at x . Such a t is a holomorphic map $X_0(2) \rightarrow \mathbb{P}^1(\mathbb{C})$ of degree 1, hence an isomorphism. $\square$

Remark 3.7. Over $\mathbb{Q}$, genus 0 alone is not enough, since a curve of genus 0 over $\mathbb{Q}$ is a conic and may have no rational points. Here, however, the cusp ∞ is a rational point of $X_0(2)$, so $X_0(2) \cong \mathbb{P}_{\mathbb{Q}}^1$ over $\mathbb{Q}$. In the next section we make this explicit.

4. RATIONAL POINTS ON $X_0(2)$

In this section we find a model for $X_0(2)$ and determine which j -invariants correspond to elliptic curves with a 2-isogeny. For this we look at the function field $\mathbb{C}(X_0(N))$.

Proposition 4.1 ([1, Proposition 7.5.1]). *The field of meromorphic functions on $X_0(N)$ is $\mathbb{C}(X_0(N)) = \mathbb{C}(j, j_N)$, where $j_N(\tau) = j(N\tau)$.*

The relationship between j and j_N is given by the modular polynomial.

Definition 4.2 ([3]). The *modular polynomial* $\Phi_N(X, Y)$ is the minimal polynomial of j_N over $\mathbb{C}(j) = \mathbb{C}(X(1))$. We may write it as

$$\Phi_N(X, j(\tau)) = \prod_{i=1}^r (X - j_N(\gamma_i \tau)),$$

where $\{\gamma_1, \dots, \gamma_r\}$ is a set of right coset representatives for $\Gamma_0(N)$ in $\mathrm{SL}_2(\mathbb{Z})$.

For $N = 2$ the classical modular polynomial is

$$\begin{aligned} \Phi_2(X, Y) = X^3 + Y^3 - X^2Y^2 + 1488(X^2Y + XY^2) - 162000(X^2 + Y^2) \\ + 40773375XY + 8748000000(X + Y) - 15746400000000. \end{aligned}$$

This polynomial is symmetric and has very large coefficients. Since $X_0(2)$ has genus 0, there is a much better model. By [corollary 3.6](#), $\mathbb{C}(X_0(2)) = \mathbb{C}(t)$ for a single function t , called a *Hauptmodul*. A standard choice is

$$t(\tau) = 2^{12} \left(\frac{\eta(2\tau)}{\eta(\tau)} \right)^{24} = 2^{12} q \prod_{n \geq 1} (1 + q^n)^{24} = 4096q + 98304q^2 + \dots,$$

where η is the Dedekind eta function and $q = e^{2\pi i \tau}$. In terms of t ,

$$(4.1) \quad j(\tau) = \frac{(t + 16)^3}{t} \quad \text{and} \quad j(2\tau) = \frac{(t + 256)^3}{t^2}.$$

So $\mathbb{C}(j, j_2) = \mathbb{C}(t)$. Writing $X = t$ and $Y = j$, the first relation in (4.1) becomes the plane model

$$F_2(X, Y) = X^3 + 48X^2 - XY + 768X + 4096 = 0,$$

and

$$\mathbb{C}(X_0(2)) \cong \mathrm{Frac}(\mathbb{C}[X, Y]/(F_2(X, Y))) = \mathbb{C}(X).$$

Remark 4.3. The map $j : X_0(2) \rightarrow X(1)$ is the rational function $X \mapsto (X + 16)^3/X$ of degree 3. Its poles are the two cusps: a simple pole at $X = 0$, which is the cusp ∞ of width 1 since $t = 4096q + \dots$, and a double pole at $X = \infty$, which is the cusp 0 of width 2. The ramification $3 = 1 + 2$ over $j = \infty$ matches [fig. 1](#).

Rational points of this model give the j -invariants we want. If $X \in \mathbb{Q} \setminus \{0\}$, then

$$Y = \frac{(X + 16)^3}{X} = \frac{X^3 + 48X^2 + 768X + 4096}{X}$$

is the j -invariant of an elliptic curve over $\mathbb{Q}$ with a 2-isogeny defined over $\mathbb{Q}$. Conversely, every such j -invariant arises this way.

Example 4.4. Take $X = -6$. Then $Y = 10^3/(-6) = -\frac{500}{3}$, so $(-6, -\frac{500}{3})$ is a point on $F_2(X, Y) = 0$. An elliptic curve with $j(E) = -\frac{500}{3}$ is

$$E : y^2 = x^3 - x^2 - 48x - 420,$$

with LMFDB label 20184.f2 [2] and $E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/2\mathbb{Z}$. The torsion is generated by the point $(10, 0)$. Since E has a rational point of order 2, it has a 2-isogeny defined over $\mathbb{Q}$, whose kernel is $\{O, (10, 0)\}$. The isogenous curve is

$$E' : y^2 = x^3 - x^2 - 1208x - 15732,$$

with LMFDB label 20184.f1, and the isogeny $\phi : E \rightarrow E'$ of degree 2 is

$$\phi(x, y) = \left(\frac{x^2 - 10x + 232}{x - 10}, \frac{y(x^2 - 20x - 132)}{(x - 10)^2} \right).$$

One way to obtain ϕ is to shift $x \mapsto x + 10$, which moves $(10, 0)$ to the origin and gives $y^2 = x(x^2 + 29x + 232)$. Applying the standard formula for a 2-isogeny with kernel generated by $(0, 0)$ then gives $y^2 = x(x^2 - 58x - 87)$, and shifting back by $x \mapsto x + 19$ gives the model of E' above.

Remark 4.5. The curve $X_0(2)$ coincides with $X_1(2)$. As groups, $\Gamma_1(2) = \Gamma_0(2)$, since the only unit modulo 2 is 1. In moduli terms, an elliptic curve E over a number field K has a 2-isogeny defined over K if and only if it has a point of order 2 defined over K . Indeed, the kernel of a K -rational 2-isogeny is a Galois-stable subgroup $\{O, P\}$, so P itself is fixed by Galois.

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